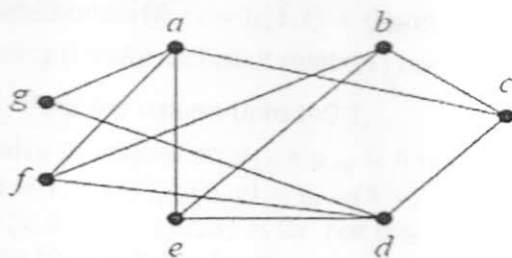


10. In Regula-Falsi method the point of intersection of the curve AB and the x - axis is replaced by
 A) point of intersection of y - axis and the curve AB
 B) point of intersection of y - axis and the chord AB
 C) point of intersection of x - axis and the chord AB
 D) point of intersection of x - axis and y - axis.
11. The root of the equation $xe^x - 3 = 0$ lies in the interval :
 A) (0,1)
 B) (1,2)
 C) (2,3)
 D) (-1,0)
12. If $y' = xy + y^2$ at $y(0) = 1$ and $k_1 = 0.1, k_2 = 0.1155, k_3 = 0.1172, k_4 = 0.13598$, then the approximate value of y at $x = 0.1$ by Runge- Kutta fourth order formula is
 A) 1.277
 B) 1.1359
 C) 1.1270
 D) 1.1168
13. For the heat equation $u_t = \frac{1}{2} u_{xx}$, $h = k = 1$, the value of mesh ratio parameter α is _____
 A) 1
 B) $\frac{1}{2}$
 C) 0.25
 D) 2
14. The forward difference approximation for $\frac{\partial u}{\partial y}$ is _____
 A) $\frac{u_{i,j} - u_{i,j-1}}{k} + O(k)$
 B) $\frac{u_{i,j+1} - u_{i,j-1}}{2k} + O(k^2)$
 C) $\frac{u_{i,j+1} - u_{i,j}}{k} + O(k)$
 D) $\frac{u_{i,j-1} - 2u_{i,j} + u_{i,j+1}}{k^2} + O(k^2)$
15. The standard five point formula for solving Poissons's equation $u_{xx} + u_{yy} = f(x, y)$ is _____
 A) $u_{i,j} = \frac{1}{4} [u_{i-1,j} + u_{i+1,j} + u_{i,j-1} + u_{i,j+1} - h^2 f(ih, jh)]$
 B) $u_{i,j} = \frac{1}{4} [u_{i-1,j} + u_{i+1,j} + u_{i,j-1} + u_{i,j+1} + h^2 f(ih, jh)]$
 C) $u_{i,j} = \frac{1}{4} [u_{i-1,j} + u_{i+1,j} + u_{i,j-1} + u_{i,j+1}]$
 D) $u_{i,j} = \frac{1}{4} [u_{i-1,j} + u_{i+1,j} + u_{i,j-1} + u_{i,j+1}] - f(ih, jh)$
16. For the wave equation $u_{tt} = u_{xx}$, $h = k = 0.2$, the value of α is _____
 A) 0.5
 B) 1.5
 C) 1.25
 D) 1
17. A "periodic function" is given by a function which:
 A) has a period $T = 2\pi$
 B) satisfies $f(t + T) = f(t)$
 C) satisfies $f(t + T) = -f(t)$
 D) has a period $T = \pi$
18. The value of a_0 in the half range Fourier cosine series expansion of $f(x) = x$ in (0,5) is
 A) 0
 B) 1
 C) 2
 D) 5
19. If $F(x(t)) = X(\omega)$, is the Fourier transform of the function $x(t)$, then $F(x(t - a)) =$
 A) $e^{-ia\omega} X(\omega)$
 B) $e^{-ia\omega} X(a\omega)$
 C) $e^{ia\omega} X(a\omega)$
 D) $\frac{e^{-ia\omega}}{X(\omega)}$
20. In the expression for DFT, $\sum_{n=0}^{N-1} x(n) e^{-i\frac{2\pi}{N}kn}$; the phase factor is the term:
 A) $x(n)$
 B) $x(n) e^{-i\frac{2\pi}{N}kn}$
 C) $e^{-i\frac{2\pi}{N}}$
 D) $\frac{2\pi}{N} kn$

PART - B: DESCRIPTIVE ANSWER QUESTIONS

	Unit - I	Marks	BT*	CO*	PO*
1. a)	(i) Define an equivalence relation. (ii) Let $A = \{0, 1, 2, 3, 4, 5\}$, and R be the relation on A defined by aRb if and only if $a \leq b + 1$. Find R and check whether R is an equivalence relation on A . Justify your answer with reason.	6	L*2	1	1

- b) Let $A = \{1, 2, 3, 4\}$ and $R = \{(1, 2), (2, 1), (2, 2), (3, 1), (4, 3)\}$. Find the transitive closure of the relation R by using Warshall's algorithm.
- c) (i) Define bipartite and complete bipartite graphs.
(ii) Check whether the following graph is bipartite. Justify your answer.



2. a) (i) Let $A = \{1, 2, 3, 4, 5, 6\}$ and $p = (1, 4) \circ (2, 3, 5)$ be a permutation of A . Compute p^{-1} and $p \circ p \circ p$. Write the result as the product of disjoint cycles.

(ii) Encode the message "GERMANY SEEKS AN ALLIANCE" using the key word "ZEBRA" by using keyword columnar transposition.

- b) With a neat diagram, explain Königsberg bridge problem.
- c) Schedule the final exams for the courses $M1, M2, M3, M4, M5, M6, M7$, using graph coloring with the fewest number of different time slots. Suppose that the following pairs of courses have common students: $M1$ and $M2, M1$ and $M3, M1$ and $M4, M1$ and $M7, M2$ and $M3, M2$ and $M4, M2$ and $M5, M2$ and $M7, M3$ and $M4, M3$ and $M6, M3$ and $M7, M4$ and $M5, M4$ and $M6, M5$ and $M6, M5$ and $M7$, and $M6$ and $M7$.

3. a) Write the following argument in symbolic form and then establish the validity of the argument using the rules of inferences.

"If Dominic goes to the racetrack, then Helen will be mad". "If Ralph plays cards all night, then Carmela gets mad". "If either Helen or Carmela gets mad, then their lawyer Veronica will be notified". "Veronica has not heard from either of these clients". Consequently, "Dominic did not make it to the racetrack and Ralph didn't play cards all night".

- b) Let $A = \{1, 2, 3, 4, 6\}$. Consider the relation R on A defined by aRb if and only if " b is a multiple of a ".

- (i) Find the relation R .
(ii) Find the matrix of R and draw the digraph of R .
(iii) Find in degree and out degree of each element of A .
(iv) List all the paths of length two from the vertex 1 in the digraph of R .

- c) Define the adjacency matrix of a graph. Also, draw the graphs for the following adjacency matrices.

$$(i) A_G = \begin{bmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{bmatrix} \quad (ii) A_H = \begin{bmatrix} 2 & 1 & 2 \\ 0 & 0 & 2 \\ 1 & 1 & 1 \end{bmatrix}$$

Unit – II

4. a) Using Modified Euler's method, find $y(0.2)$ given that $\frac{dy}{dx} = y + e^x$, $y(0) = 0$, take $h = 0.2$, carry out 4 iterations.

6 L3 1 2

4 L2 2 1

6 L2 1 2

6 L2 2 2

4 L3 2 1

6 L3 1 2

6 L2 1 1

4 L2 2 2

6 L3 3 1

- b) Solve the Poisson's equation $\nabla^2 u = x^2 + y^2$, $0 < x < 3$, $0 < y < 3$ given that $u(x, y) = 0$ on the boundary, take $h = k = 1$. 6 L3 4 2
- c) Find a real root of the equation $xe^x - 1 = 0$, using Newton-Raphson method near $x_0 = 1$. Carry out two iterations. 4 L2 3 1
5. a) Solve the boundary value problem $u_t = u_{xx}$ under the conditions $u(0, t) = u(1, t) = 0$ and $u(x, 0) = \sin \pi x$, $0 \leq x \leq 1$ using Bendre-Schmidt relation (take $h = 0.2$, $k = 0.02$ and $\alpha = \frac{1}{2}$). Find the values upto $t=0.1$. 6 L3 4 2
- b) Solve the equation $u_{xx} + u_{yy} = 0$ under the conditions ($h = 1, k = 1$), $u(0, y) = 0$, $u(3, y) = 3 + y$, for $0 \leq y \leq 3$; $u(x, 0) = x$, $u(x, 3) = 2x$ for $0 \leq x \leq 3$. 6 L3 4 1
- c) Use Runge-Kutta fourth order formula to find $y(0.1)$, given that $y' = x + y^2$, $y(0) = 1$, take $h = 0.1$. 4 L2 3 1
6. a) Find a real root of the equation $x^3 - 2x - 5 = 0$ in the interval $(2, 3)$, using regula-falsi method. Carry out three iterations. 6 L2 3 1
- b) Evaluate the pivotal values of the equation $u_{tt} = 16u_{xx}$, taking $h = 1$ upto $t = 1.25$ and $k = \frac{1}{4}$. The boundary conditions are $u(0, t) = u(5, t) = 0$, $u_t(x, 0) = 0$ and $u(x, 0) = x^2(5 - x)$. 6 L3 4 2
- c) Use Taylor's series method to find the value of y at $x = 0.1$, given that $\frac{dy}{dx} = x + y$, $y(0) = 1$, take $h = 0.1$. 4 L3 3 1

Unit - III

7. a) Find the Fourier series expansion of the function $f(x) = \begin{cases} 0 & -2 < x < 0 \\ 1 & 0 < x < 2 \end{cases}$ and $f(x + 4) = f(x)$. 6 L2 5 1
- b) Show that $F_S\{t f(t)\} = -\frac{d}{d\omega}\{F_C(\omega)\}$; $F_C\{t f(t)\} = \frac{d}{d\omega}\{F_S(\omega)\}$. Hence find the Fourier sine transform of te^{-at} . 6 L3 5 2
- c) Express $f(x) = \frac{x}{2}$ as a Fourier series in the interval $-\pi < x < \pi$. 4 L3 5 1
8. a) Find the Fourier transform of $f(t) = \begin{cases} 1 & \text{for } |t| < 1 \\ 0 & \text{for } |t| > 1 \end{cases}$. Hence evaluate $\int_0^\infty \frac{\sin t}{t} dt$. 6 L3 5 1
- b) Find the N - point DFT of the following signals:
(i) $x(n) = e^{j(\frac{2\pi}{N})k_0 n}$; $0 \leq n \leq N - 1$
(ii) $y(n) = e^{-j(\frac{2\pi}{N})k_0 n}$; $0 \leq n \leq N - 1$. 6 L3 5 2
- c) Find the half range Fourier cosine series of $f(x) = e^x$ in $(0, \pi)$. 4 L2 5 2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
